

Experiment:-17

Problem Statement:-

In Linux, the `wc` command is commonly used to count lines, words, and characters in a file. However, understanding how to perform these operations manually using shell scripting helps students gain deeper insight into text processing, looping constructs, and command substitution. This knowledge is particularly useful in automation, log analysis, and cyber security scripting tasks.

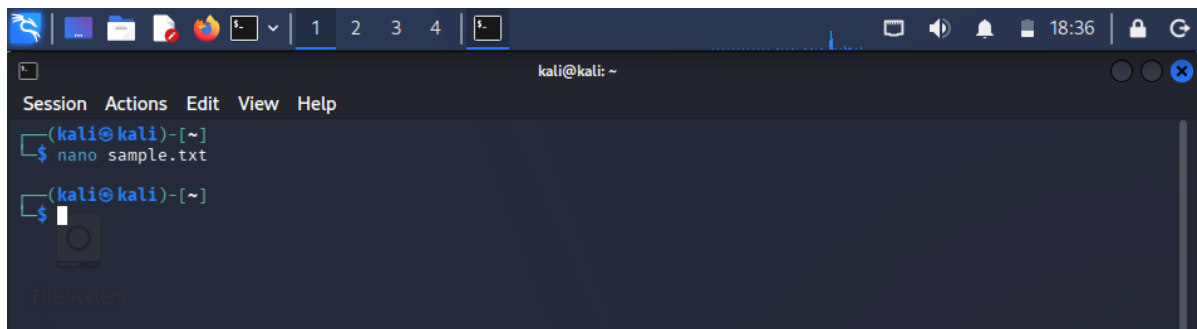
The problem is to write a shell script that counts the number of lines, words, and characters in a given input file without using the `wc` command, using Bash scripting techniques and basic text-processing logic.

Aim:-Shell script to count line, word and character without using `wc`

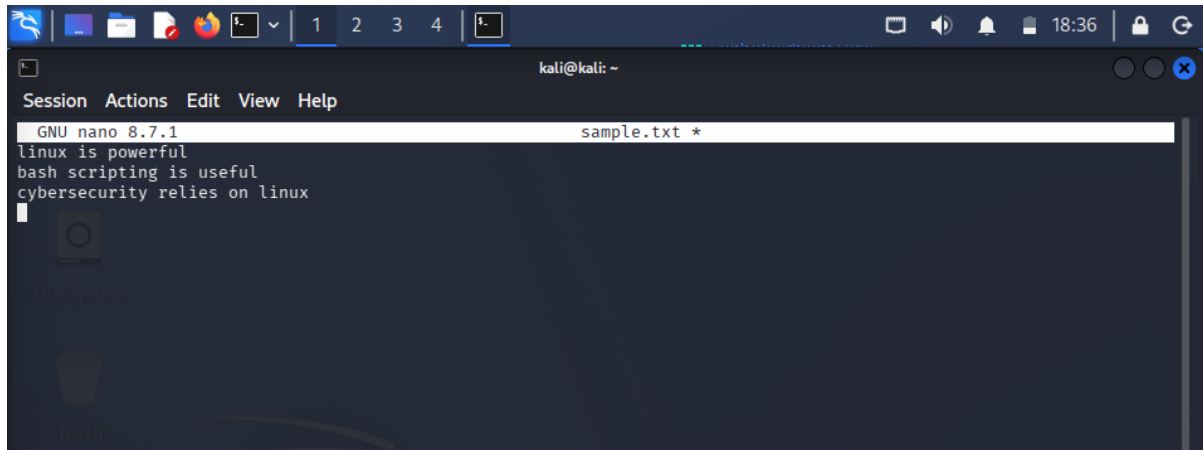
Procedure:-

A. Creating Sample Input File

1. Open the terminal.
2. Create a sample text file: `nano sample.txt`



3. Enter the following content: Linux is powerful ,Bash scripting is useful ,Cyber Security relies on Linux



The screenshot shows a terminal window with a dark background. At the top, there's a window title bar with icons and the text 'kali@kali: ~'. Below it is a menu bar with 'Session', 'Actions', 'Edit', 'View', and 'Help'. The main area shows the 'GNU nano 8.7.1' editor interface. The file being edited is 'sample.txt *'. The content of the file is: 'linux is powerful', 'bash scripting is useful', and 'cybersecurity relies on linux'. The cursor is at the end of the third line.

4. Save and exit the editor.

B. Writing the Shell Script

5. Create a script file: nano count1_text.sh



The screenshot shows a terminal window with a dark background. At the top, there's a window title bar with icons and the text 'kali@kali: ~'. Below it is a menu bar with 'File', 'Actions', 'Edit', 'View', and 'Help'. The main area shows the 'GNU nano 8.7.1' editor interface. The file being edited is 'sample1_txt.sh *'. The content of the file is: '(srishti_m@kali)-[~]', '\$ nano sample.txt', '(srishti_m@kali)-[~]', '\$ nano count1_text.sh', and '(srishti_m@kali)-[~]'. The cursor is at the end of the fifth line.

6. Enter the following script:

```
(kali㉿kali)-[~]  
$ chmod +x count1_text.sh
```

```
(kali㉿kali)-[~]  
$
```

File Actions Edit View Help

GNU nano 8.4

count1_text.sh

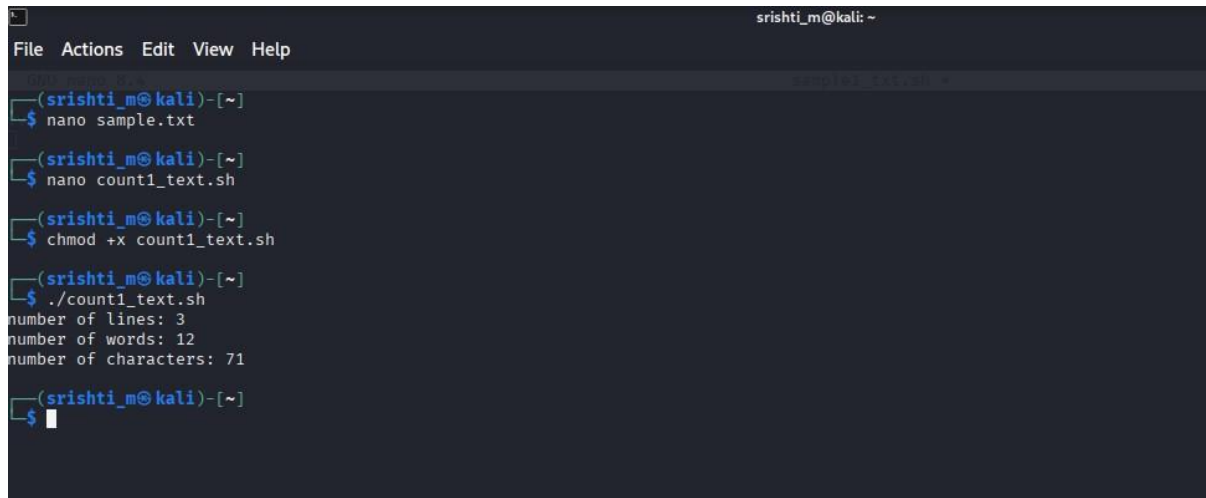
```
#!/bin/bash  
lines=0  
words=0  
chars=0  
while read line  
do  
    lines=$((lines + 1))  
    chars=$((chars + ${#line}))  
    for word in $line  
    do  
        words=$((words + 1))  
    done  
done < sample.txt  
echo "number of lines: $lines"  
echo "number of words: $words"  
echo "number of characters: $chars"
```

7. Save and exit the editor.

8. Make the script executable: `chmod +x count1_text.sh`

C. Executing the Script

9. Run the script: `./count1_text.sh`

A terminal window with a dark background and light blue text. The window title is 'srishti_m@kali: ~'. The menu bar shows 'File', 'Actions', 'Edit', 'View', and 'Help'. The terminal shows a sequence of commands and their outputs. The first command is 'nano sample.txt', followed by 'nano count1_text.sh', and then 'chmod +x count1_text.sh'. The final command is './count1_text.sh', which outputs three lines: 'number of lines: 3', 'number of words: 12', and 'number of characters: 71'. The prompt '\$' is visible at the end of the last command line.

```
srishti_m@kali: ~  
File Actions Edit View Help  
srishti_m@kali: ~  
$ nano sample.txt  
$ nano count1_text.sh  
$ chmod +x count1_text.sh  
$ ./count1_text.sh  
number of lines: 3  
number of words: 12  
number of characters: 71  
$
```